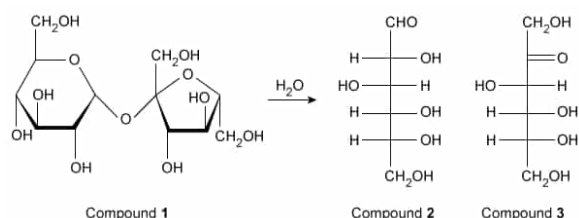


Shown below is the hydrolysis of a particular disaccharide.



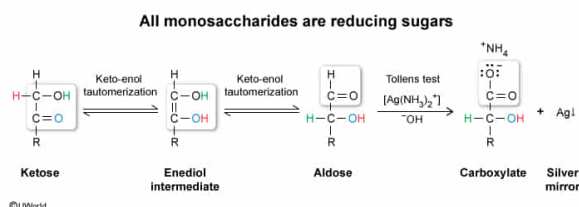
Which of the carbohydrates will be oxidized by the Tollens reagent (silver mirror test) to a carboxylic acid?

- ☐ A. Compound 1 only [7%]
- ☐ B. Compound 2 only [46%]
- ✓ ☒ C. Compounds 2 and 3 only [32%]
- ☐ D. Compounds 1, 2, and 3 [13%]

Correct

33% answered correctly

## Explanation:



The Tollens test (silver mirror test) is a qualitative test that detects **reducing sugars**. These sugars can reduce the Tollens and other reagents and, as a consequence, are themselves *oxidized*. Structurally, reducing sugars can be recognized by having a free **anomeric carbon**. In cyclic sugars, anomeric carbons can be distinguished as free if at least one of the attached oxygens is part of a hydroxyl group, thereby showing that particular anomeric carbon is *not* involved in a **glycosidic bond**.

During the oxidation of reducing sugars, cyclic sugars are linearized first (if possible) and then the carbonyls of linear sugars are **oxidized**. The aldehydes of **aldoses** can be directly oxidized to carboxylic acids; however, the ketones of **ketoses** must undergo **keto–enol tautomerization** before the sugar can be oxidized. During tautomerization, the ketone on carbon 2 of the ketose tautomerizes to form a double bond between carbon 1 and carbon 2. Because both carbon 1 and carbon 2 have alcohols, this intermediate is known as an **enediol**. Tautomerization of the enediol can reform a carbonyl (this time on carbon 1) to form the aldehyde of an aldose sugar, which can then be oxidized.

In the reaction shown, the products of the hydrolysis of a disaccharide are monosaccharides. The glycosidic bond connecting them has been broken, ensuring that both anomeric carbons of the products are now free, forming an aldose and a ketose,

respectively. Due to tautomerization, both aldoses and ketoses can be oxidized, and thus all monosaccharides are reducing sugars.

**(Choices A and D)** Compound **1** is a disaccharide. Upon examining the anomeric carbons of the constituent monosaccharides, neither anomeric carbon is **free**, indicating that Compound **1** is nonreducing.

**(Choice B)** Compound **3** is also a reducing sugar due to tautomerization, and will result in a positive Tollens test.

**Educational objective:**

All monosaccharides, both aldoses and ketoses, are considered reducing sugars. Disaccharides and other oligosaccharides are considered reducing only if one of the anomeric carbons is free (ie, not involved in a glycosidic bond).

**Time spent:**0 sec

**Subject:**

Biochemistry

**QID:**403502

**System:**

Carbohydrates, Nucleotides, and Lipids